

Ananya Joshi

ananyajoshi100@gmail.com ❖ (609) 865-3175 ❖ New York, NY

WORK EXPERIENCE

Commodity and Ingredient Hedging – Software Engineer

CIH provides commodity risk management and hedging solutions to 10,000+ agricultural and financial clients.

August 2023 - Present

Platform Modernization and Architecture [TypeScript/React]

- Led the architectural pitch and ongoing migration from legacy CFML to a decoupled React/RESTful stack, introducing a modular UI library and standardized backend repository patterns.
- Built reusable React hooks and data-fetching abstractions (TanStack Query) to standardize form state and API interactions, reducing frontend boilerplate by 30% and accelerating feature delivery.
- Delivered the first end-to-end implementation of the new architecture, defining strongly-typed request/entity models, centralized error handling, and scalable patterns for complex filtering and CRUD workflows.

Distributed Call Processing System [Python/AWS/SQL]

- Built a distributed background processing system handling 10k+ daily calls, orchestrating ingestion, status polling, S3 transfers, and cold-storage archiving across horizontally scalable workers.
- Designed an automated reconciliation system auditing 10k+ daily records, guaranteeing end-to-end consistency between third-party providers and internal storage.
- Built a transcript processing layer within the pipeline using AWS Bedrock to transform archived call data into structured, searchable summaries, reducing manual note-taking effort by 80%.
- Delivered a React/TypeScript dashboard in the new architecture, enabling real-time filtering, participant management, and editing of AI-generated summaries.

Internal Analytics for Business Intelligence [SQL/Lucee]

- Designed and built data transformation pipelines to aggregate disparate business metrics (orders, communications, alerts) into a centralized KPI system used by 12 internal product teams for performance tracking and decision-making.

Revenue Growth Tooling [React]

- Built an interactive “what-if” revenue modeling tool for client profiles, enabling sales teams to simulate product scenarios and identify upsell opportunities through real-time margin comparisons.

GAP – Software Engineer Intern

Global apparel and retail company serving millions worldwide through e-commerce and 3,000+ stores.

May 2022 - August 2022

Internal Pricing and Promotion Validator [React/Java]

- Identified a bottleneck in manual promotion verification; built a functional web-based prototype to automate the audit process, interfacing with internal REST APIs to fetch real-time retail data.
- Designed wireframes and implemented a custom React interface to cross-reference real-time pricing data against internal promotional rules to ensure accuracy during the validation cycle.
- Partnered with two engineers to deliver the client-side implementation, successfully demonstrating a 30% reduction in verification time during the pilot phase.
- Developed a JUnit test suite to validate secure data handling and functional logic for credit card information storage modules.

EDUCATION

University of Michigan – Ann Arbor, MI

Bachelor of Science in Computer Science, School of Engineering

September 2019 – May 2023

TECHNICAL SKILLS

- *Languages:* JavaScript/TypeScript, Python, C++, CFML (Lucee)
- *Frontend:* React, HTML, CSS
- *Backend and APIs:* Node.js, Express, REST APIs, Taffy
- *Databases:* MySQL, MongoDB, Firebase, Google BigQuery
- *Cloud and Tools:* AWS (S3, Bedrock), Docker, Git, Bash scripting